

Project Report

Risk-Sensitive Reinforcement Learning for Study Scheduling: A CVaR-DQN Approach

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Abstract

We address optimal exam study scheduling using reinforcement learning. A student must allocate limited study days across multiple subjects, balancing knowledge acquisition against forgetting, energy constraints, and subject dependencies. Standard DQN maximizes expected total score but may sacrifice individual exams. We apply **CVaR-DQN**, a distributional RL method that optimizes the Conditional Value at Risk, to produce risk-sensitive schedules that protect against worst-case outcomes. We evaluate seven methods across three environment configurations: CVaR-DQN consistently improves CVaR_{10%} over DQN while sacrificing a small amount of mean return, exactly the trade-off the algorithm is designed to make. Ablation studies show energy stochasticity is the primary source of outcome variance, and removing it eliminates the mean–risk gap entirely. As a pilot, we explore *Subject-Constrained CVaR-DQN*—a Multi-Objective Distributional Constrained MDP with per-subject CVaR constraints—which reduces Physics failure on Medium from 30% to 8% but underperforms on aggregate metrics and under distribution shift, motivating adaptive thresholding as future work.

1 Introduction

Students face a sequential decision problem when preparing for exams: which subject to study each day, given limited time, stochastic cognitive energy, and exponential forgetting. Standard reinforcement-learning agents maximize expected total score, which can lead to policies that sacrifice low-weight exams to inflate the aggregate. In practice, failing even one exam carries severe consequences. We apply **CVaR-DQN**—a distributional RL method that optimizes the Conditional Value at Risk [Chow et al., 2015, Dabney et al., 2018]—to produce risk-sensitive study schedules that protect against worst-case outcomes, and additionally explore an extension that imposes per-subject constraints.¹

Related work. RL has been applied to flashcard review timing [Reddy et al., 2017] and broader educational sequencing [Doroudi et al., 2019]. We build on DQN [Mnih et al., 2015] and Double DQN [Van Hasselt et al., 2016], and extend QR-DQN [Dabney et al., 2018] with CVaR action selection [Chow et al., 2015]. Our pilot extension combines CMDP-style constrained optimization [Altman, 1999, Achiam et al., 2017] with distributional RL by constraining per-subject CVaR.

Contributions. (i) A Gymnasium environment modeling exam scheduling with forgetting, energy, and prerequisites; (ii) a comparison of seven scheduling methods across three environment scales with ablation and robustness analyses; (iii) empirical evidence that CVaR-DQN consistently improves worst-case CVaR_{10%} over DQN at a small cost in mean return; (iv) a pilot study of *Subject-Constrained CVaR-DQN*, a Multi-Objective Distributional Constrained MDP formulation, with positive and negative findings.

2 Preliminaries

We model exam preparation as a finite-horizon MDP. The **state** is $s = (\mathbf{k}, e, d)$ where $\mathbf{k} \in \{0, \dots, K\}^N$ is the knowledge vector across N subjects, $e \in \{1, \dots, E\}$ is energy, and $d \in \{1, \dots, D\}$ is the current day. The **action** space is $a \in \{0, \dots, N\}$: study subject i or rest ($a = N$). Dynamics:

$$\text{Forgetting:} \quad k_i \leftarrow \max(0, \lfloor k_i \cdot e^{-\lambda_i} \rfloor) \quad \forall i \neq a \quad (1)$$

$$\text{Learning:} \quad k_a \leftarrow \min(K, k_a + \lfloor \frac{e}{E} g_{\max}(1 + \text{bonus}(a)) \rfloor) \quad (2)$$

$$\text{Energy:} \quad e \leftarrow \text{clip}(1, E, e - c_{\text{study}} + 1 + c_{\text{rest}} + \xi) \quad (3)$$

where $\lambda_i \in [0.05, 0.2]$ is the subject-specific forgetting rate, $\text{bonus}(a) = \alpha \min_{p \in \text{prereq}(a)} k_p / K$ when all prerequisites exceed threshold β (zero otherwise), and $\xi \sim \text{Uniform}\{-1, 0, +1\}$ is the energy noise. The **reward** is $r_{\text{step}} = -0.1$ per step plus an exam-day reward $r_{\text{exam}} = w_i \sigma(k_i - K/2)$ on

¹Code and experiment scripts: <https://github.com/muhammadv-q/cvar-dqn-exam-scheduler>

subject i 's exam day, where w_i is its credit weight and σ is the sigmoid. The sigmoid maps zero knowledge to near-zero reward, $K/2$ knowledge to $0.5w_i$, and mastery to w_i .

Configurations. Three configurations of increasing scale: **Small** ($N=3$, $K=5$, $D=7$; ~ 7.6 k states), **Medium** ($N=4$, $K=8$, $D=10$; ~ 328 k states), and **Large** ($N=5$, $K=10$, $D=14$; ~ 11.3 M states). All use $E = 5$.

3 Methods

Baselines. Three non-learning heuristics—**Uniform** (round-robin), **Most-Urgent-First** (next exam first), **Lowest-Knowledge-First**—and three learning baselines: **Value Iteration** (optimal on Small), **tabular ϵ -greedy Q-Learning** (Small/Medium), and **DQN with Double DQN** [Mnih et al., 2015, Van Hasselt et al., 2016], a two-layer MLP (128 units, ReLU) with experience replay and target network.

CVaR-DQN. The main contribution. CVaR-DQN [Dabney et al., 2018] learns $M = 32$ quantile values $\{\theta_i(s, a)\}$ per action and selects actions by

$$a^* = \arg \max_a \text{CVaR}_\alpha(s, a), \quad \text{CVaR}_\alpha(s, a) = \frac{1}{\lceil \alpha M \rceil} \sum_{i=1}^{\lceil \alpha M \rceil} \theta_{(i)}(s, a), \quad (4)$$

with $\alpha = 0.1$ (worst 10%). Training uses the quantile Huber loss [Dabney et al., 2018] with asymmetric weighting that pushes each quantile toward its target position in the distribution. Where DQN maximizes the *average* outcome, CVaR-DQN maximizes the *worst-case* outcome.

Subject-Constrained CVaR-DQN (pilot). An extension that treats exam scheduling as a Multi-Objective Distributional Constrained MDP. Let $Z_i(s, a)$ be the return attributable to subject i and define a passing threshold $T_i = \frac{1}{2}w_i$ (sigmoid midpoint, $\equiv$ knowing $K/2$). The ideal problem $\max_a \sum_i \mathbb{E}[Z_i(s, a)]$ s.t. $\text{CVaR}_\alpha(Z_i(s, a)) \geq T_i \forall i$ is impractical to solve directly (often-empty feasible set destabilizes Bellman targets), so we use the Lagrangian relaxation

$$a^* = \arg \max_a \sum_i \mathbb{E}[Z_i(s, a)] - \lambda \sum_i [\max(0, T_i - \text{CVaR}_\alpha(Z_i(s, a)))]^2. \quad (5)$$

with $\lambda = 5$. The quantile network outputs $N \times M$ values per action (one distribution per subject); the quantile Huber loss is averaged across subjects, with per-step reward r_i being the exam score for subject i .

4 Experiments and Results

Setup. All RL agents are trained for 5,000 episodes and evaluated over 1,000 episodes across 5 seeds (42, 123, 456, 789, 1024). Metrics: **Mean Return** (average total reward), **CVaR_{10%}** (mean of worst 10% returns), **Mean Worst Exam** (lowest individual exam score per episode).

4.1 Main Results

Table 1: Performance comparison across all configurations (mean over 5 seeds). Bold marks the best agent within each config for CVaR_{10%}.

Config	Agent	Mean Return	CVaR _{10%}	Worst Exam
Small	Uniform	0.68	0.03	0.20
	MUF	3.56	2.55	0.51
	LKF	0.76	0.22	0.15
	Value Iteration	5.00	3.44	0.15
	Q-Learning	2.85	1.80	0.27
	DQN	4.97	3.37	0.15
	CVaR-DQN	4.85	3.48	0.15
	SC-CVaR-DQN	4.68	3.23	0.22
Medium	Uniform	−0.71	−0.78	0.04
	MUF	3.11	1.38	0.24
	LKF	−0.48	−0.72	0.04
	Q-Learning	1.47	0.59	0.21
	DQN	4.86	2.86	0.04
	CVaR-DQN	4.50	3.19	0.04
	SC-CVaR-DQN	4.53	2.40	0.04
Large	Uniform	−1.05	−1.29	0.02
	MUF	8.50	2.25	0.14
	LKF	−0.88	−1.25	0.02
	DQN	11.47	8.25	0.01
	CVaR-DQN	10.62	8.27	0.01
	SC-CVaR-DQN	11.19	8.10	0.01

Key findings. DQN achieves the highest mean return across all configurations, closely matching Value Iteration on Small (4.97 vs. 5.00). CVaR-DQN consistently achieves higher CVaR_{10%}, confirming more robust schedules: on Medium, DQN gets 4.86 mean vs. CVaR-DQN’s 4.50 (−7% mean) but CVaR_{10%} flips to 2.86 vs. 3.19 (+12% worst-case)—the mean-risk trade-off the algorithm is designed to make. SC-CVaR-DQN is competitive on mean return but underperforms on aggregate CVaR; we analyze its specific behavior in §4.4. Among heuristics, Most-Urgent-First performs best. All three deep RL agents converge within ∼2,000 episodes (Appendix Figure 2); a representative-config visualization (Medium) is in Figure ?? (appendix).

4.2 Ablation Study

We ablate one component at a time on the medium configuration (Table 2, visual in appendix Figure 5).

Table 2: Ablation results (medium config, mean over 5 seeds). Bold values mark the better agent within each row for CVaR_{10%}.

Ablation	Mean Return		CVaR _{10%}	
	DQN	CVaR-DQN	DQN	CVaR-DQN
Baseline	4.79	4.50	2.91	3.19
No forgetting	5.63	5.59	4.34	4.95
Constant energy	7.01	7.01	7.01	7.01
No prerequisites	4.86	4.58	2.91	3.17
Equal weights	0.82	0.74	0.33	0.31
$\lambda = 0.05$	4.85	4.52	2.88	3.13
$\lambda = 0.2$	4.83	4.54	2.96	3.14
$\lambda = 0.4$	4.57	4.36	3.03	2.86

Constant energy eliminates all variance: both agents achieve the deterministic optimum (7.01)—when energy noise is removed there is no risk to be sensitive to, confirming energy stochasticity is the sole source of uncertainty differentiating the two. **No forgetting** boosts both agents and grows CVaR-DQN’s advantage (4.95 vs. 4.34). **High forgetting** ($\lambda = 0.4$) is the only condition where DQN’s CVaR exceeds CVaR-DQN’s—rapid decay rewards aggressive late study over caution. **Prerequisites** have minimal effect.

4.3 Robustness Tests

We evaluate agents trained on normal conditions under three perturbations (Table 3, appendix Figure 6).

Table 3: Robustness results (mean over 5 seeds). Bold values mark the best agent within each row for CVaR_{10%}.

Test	Mean Return			CVaR _{10%}		
	DQN	CVaR-DQN	SC-CVaR-DQN	DQN	CVaR-DQN	SC-CVaR-DQN
Normal	4.86	4.50	4.53	2.90	3.19	2.40
Wide energy noise	4.08	3.79	3.69	1.56	1.55	1.13
Compressed exams	2.76	2.92	1.89	0.80	1.34	0.30
High variance	4.07	3.56	3.52	1.54	1.27	1.09

Under **compressed exams**, CVaR-DQN maintains the highest CVaR_{10%} (1.34 vs. 0.80 for DQN)—its conservative strategy transfers naturally when exam clustering removes recovery slack. Under **wide energy noise** and **high variance** (doubled forgetting + wide noise), all three RL agents degrade; DQN retains slightly higher CVaR. **SC-CVaR-DQN does worse than both baselines under all perturbations**—the constraint thresholds T_i are tuned to the training distribution, so the agent over-hedges against the wrong type of risk under shift.

4.4 Pilot: Subject-Constrained CVaR-DQN

We examine per-subject failures: Table 4 reports the fraction of episodes in which each subject’s exam score falls below $T_i = 0.5w_i$.

Table 4: Per-subject failure rate (fraction of episodes with exam score $< 0.5w_i$, mean over 5 seeds). S_i = subject i ; bold marks the lowest failure rate per subject among the three RL agents. “—” indicates the subject does not exist in this configuration.

Config	Agent	S_0	S_1	S_2	S_3	S_4
Small	Value Iteration	0.00	0.08	1.00	—	—
	DQN	0.00	0.10	1.00	—	—
	CVaR-DQN	0.11	0.11	1.00	—	—
	SC-CVaR-DQN	0.02	0.16	1.00	—	—
Medium	DQN	0.04	0.30	1.00	0.50	—
	CVaR-DQN	0.40	0.46	1.00	0.14	—
	SC-CVaR-DQN	0.04	0.08	1.00	0.53	—
Large	DQN	0.00	0.01	0.04	1.00	0.12
	CVaR-DQN	0.20	0.20	0.01	1.00	0.04
	SC-CVaR-DQN	0.00	0.05	0.02	1.00	0.13

Diagnostic. On every configuration, at least one subject fails 100% of the time—and crucially, this holds *even for Value Iteration*, the DP optimum. The lowest-weight subject ($w = 2$: Subject 2 on Small/Medium, Subject 3 on Large) has maximum exam reward ≈ 1.85 , which is smaller than the marginal reward of spending those days on higher-weight subjects; the reward-optimal policy rationally sacrifices it.

What SC-CVaR-DQN achieves. The one positive: **on Medium, SC-CVaR-DQN reduces Physics failure (S_1) from 30% (DQN) and 46% (CVaR-DQN) to 8%**—a 4–6 $\times$ reduction—while Math failure stays low (0.04). This is the algorithm working as intended: the per-subject distributional forecast and Lagrangian penalty reallocate effort toward an at-risk-but-protectable subject.

What it does *not* achieve. (i) Cannot protect environment-infeasible subjects: increasing λ from 5 to 100 makes no difference. (ii) Underperforms on aggregate metrics: mean return below DQN’s by ~ 0.3 on every config; $\text{CVaR}_{10\%}$ below both DQN and CVaR-DQN on Medium (2.40 vs 2.86 vs 3.19). (iii) Degrades under distribution shift (Table 3)—thresholds tuned to the training distribution misfire when test conditions change. On a deliberately easy “Fair” configuration (equal weights, no forgetting, no prerequisites), DQN/CVaR-DQN reach 0% any-failure while SC-CVaR-DQN sits at 2–3%—the framework permits no-failing, and the algorithm is a small cost when no constraint actually binds.

5 Conclusion

We presented a risk-sensitive RL approach to exam study scheduling. CVaR-DQN consistently produces higher $\text{CVaR}_{10\%}$ than DQN across all three configurations, at a small cost in mean return—the mean-risk trade-off in finance. Ablations show that energy stochasticity is the dominant source of outcome variance (removing it makes DQN and CVaR-DQN identical), while forgetting affects difficulty level uniformly. The SC-CVaR-DQN pilot validates the formulation (per-subject distributional constraints are computable and trainable) and yields a clear positive (4–6 $\times$ reduction in Physics failure on Medium) embedded in clear negatives (lower aggregate metrics, worse robustness, no help on environment-infeasible subjects). The root cause of the negatives is fixed thresholds—they neither adapt to environment dynamics nor to test-time distribution—and adaptive thresholding is the natural follow-up direction.

Limitations & future work. The environment uses discrete knowledge levels and a uniform energy-noise model; real cognitive energy is autocorrelated and influenced by exogenous factors. A natural next step is to fit the energy and forgetting processes to *real student data* (sleep-tracker logs, learning-platform telemetry) so the agent can be trained on empirically grounded dynamics. SC-CVaR-DQN training is $\sim 1.3\times$ slower than CVaR-DQN due to the N -fold quantile output.

References

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A Supplementary Figures

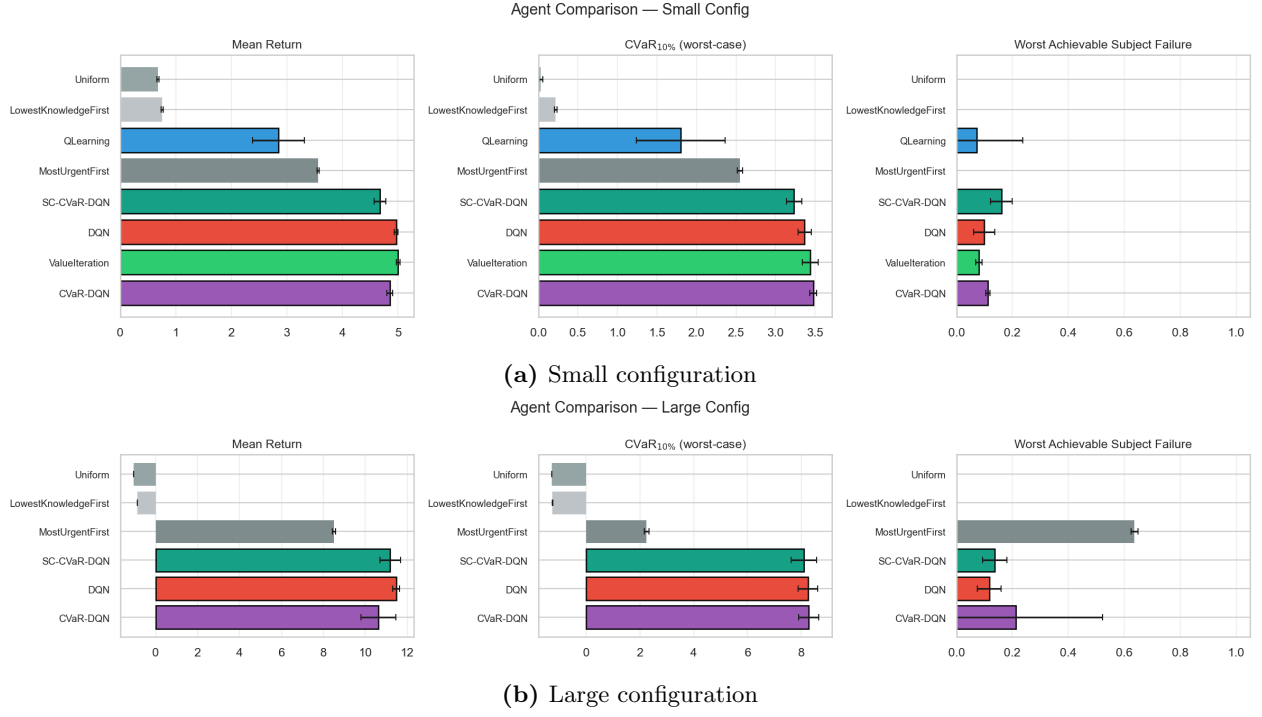


Figure 1: Agent comparison on Small and Large configurations (the Medium-config version appears in the main text as Figure ??).

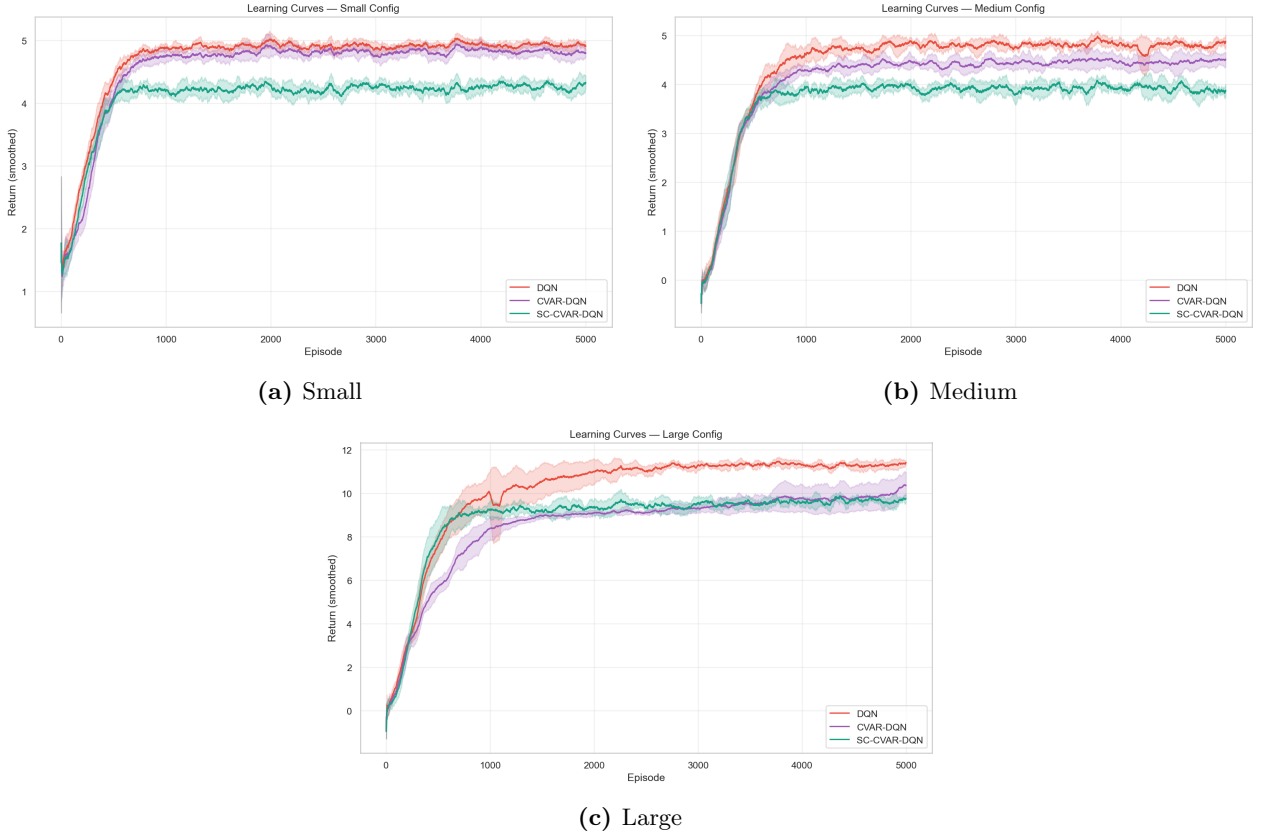


Figure 2: Learning curves (mean $\pm$ std over 5 seeds) for DQN, CVaR-DQN, and SC-CVaR-DQN.

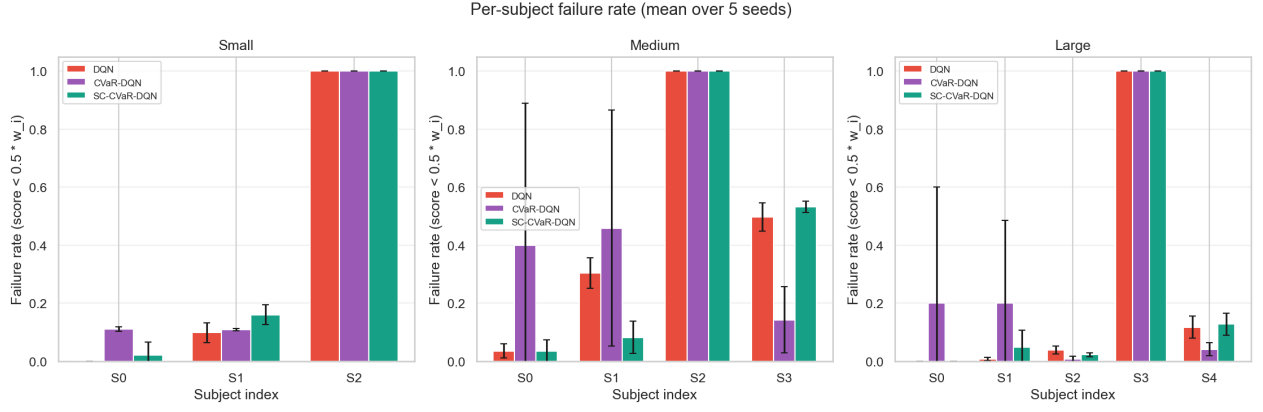


Figure 3: Per-subject failure rate (mean $\pm$ std over 5 seeds), DQN vs. CVaR-DQN vs. SC-CVaR-DQN. Same data as Table 4.

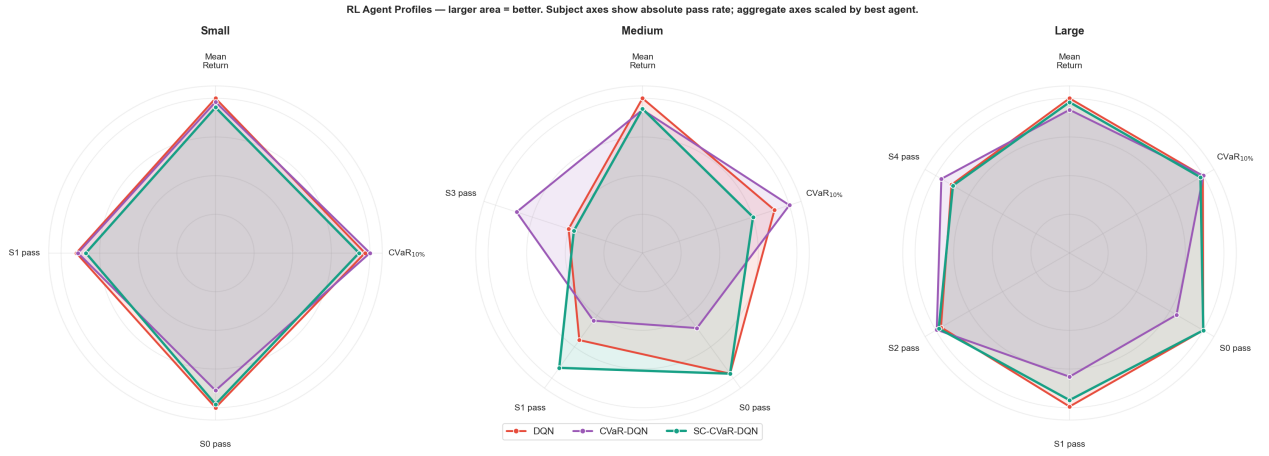


Figure 4: Multi-dimensional profile of each RL agent per configuration. Subject axes show absolute pass rate (1– failure rate); aggregate axes (Mean Return, CVaR_{10%}) are scaled so the best agent on each axis sits at 1. Universally-infeasible subjects are omitted. Larger enclosed area = better.

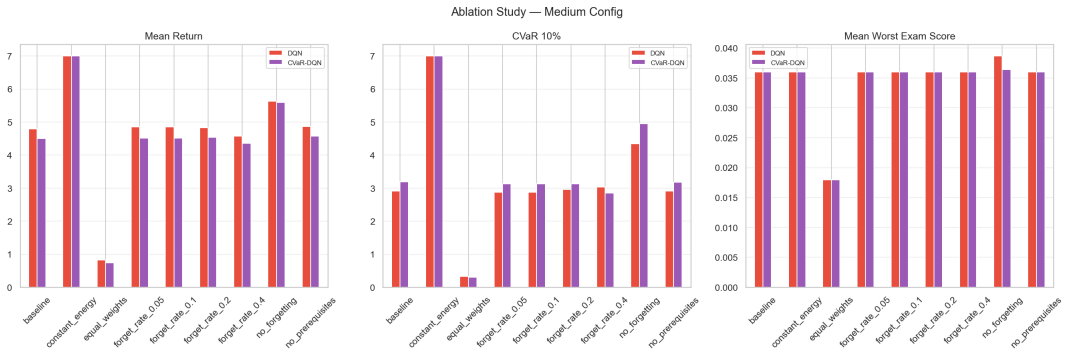


Figure 5: Ablation study visual summary; same data as Table 2.

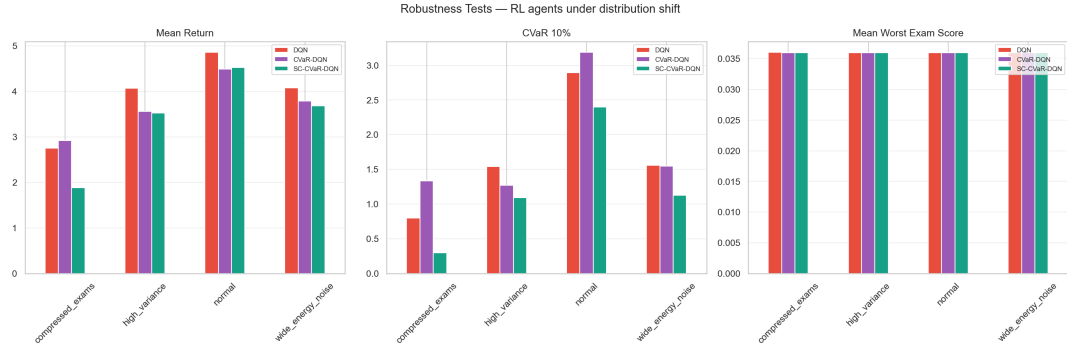


Figure 6: Robustness tests visual summary; same data as Table 3.